Tudor Andrei Diac

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Research Interests

Machine learning grounded in computational mathematics, numerical methods, and statistical modelling, with an emphasis on small, interpretable, and calibrated models and on quantifying what learned representations actually capture; representation and latent-prediction methods for dynamical and physical systems; efficient, data-centric pipelines and edge-ready deployment of ML systems.

Education

Lund University, MSc in Computational Science (Scientific Computing) Sep 2024 – June 2026

- **Thesis:** *Pretraining, Quantization, and Evaluation of Romanian Small Language Models*, graded VG (Pass with distinction) - Open access.

University of Bucharest, BSc in Physics Informatics Oct 2021 – July 2024

- **Thesis:** *The Study of the Production of Charm Mesons and Tau Neutrinos in the DsTau Experiment* (CERN collaboration), graded 10/10

Research Experience

Research Collaborator, Institute of Space Science (ISS), Romania May 2025 – June 2026

- Led a project on Romanian small language models: built a 21.16 GB curated corpus (approx. 5.6B tokens) from web crawls, targeted scrapers and literary texts, and pretrained four 300M-parameter variants from scratch (multi-head, grouped-query, multi-query and sliding-window attention)
- Best model achieved the lowest perplexity on 4 of 5 held-out Romanian domains among available models, including models with up to 10 times more parameters; all four variants surpassed every published Romanian baseline on LaRoSeDa sentiment classification, including RoBERT-large
- 4-bit quantization retained over 98.5% of full-precision fluency at under 200 MB per model, enabling on-device inference on NVIDIA Jetson and Raspberry Pi
- Contributed to research on low-resource language model evaluation and on authorship verification using frequency and structural statistics

Research Technician III, DsTau (NA65) CERN Project, ISS, Romania Apr 2024 – Sep 2024

- Built a machine-learning particle-momentum estimator and integrated its outputs into the DsTau (NA65) analysis pipeline
- Built a neural-network ensemble to identify $D_s \to \tau$ candidate decays, reaching 94.2% accuracy on a held-out test set and replacing manual screening with automated selection

Publications

T.A. Diac, P.F. de Viana, A.F. Neagoe, A. Oprea, M.C. Raportaru, A. Nicolin-Żaczek.
Edge-Ready Romanian Language Models: Training, Quantization, and Deployment.
AI (MDPI), 2026. doi:10.3390/ai7020061

A. Nicolin-Żaczek, P.-A. Gogîță, T.-G. Dumitru, F.-I. Constantin, **T.A. Diac**, A. Neagoe, M.-C. Raportaru.
Scale-free to Pareto-Tsallis transitions in the distributions of waiting times: weather, sea-level, currency trading and automotive datasets.
Journal of Physics: Complexity, 2026. doi:10.1088/2632-072X/ae8aa5

T.A. Diac, A.F. Neagoe, M.C. Raportaru, A. Tolea, R.-M. Drăgan, A. Nicolin-Żaczek.
Statistics of a large-scale Romanian corpus for language modelling.
Romanian Journal of Physics, 2025. doi:10.59277/RomJPhys.2025.70.111

Research Projects

Identifiability of Cross-Modal Correspondence under Weak Supervision 2026 – present

- Formulated region-phrase correspondence in image-caption data as a latent many-to-many coupling that is observed only through bag-level pairing, separating what each modality observes from what the pairing reveals
- Measured with CLIP on Flickr30k Entities that the true region–phrase correspondence is substantially richer than the one-to-one pairing assumed by contrastive training
- Built a synthetic testbed with known ground-truth coupling in which recovery succeeds under a support-variability condition on component co-occurrence and fails precisely where it is violated; modelled modality-private parts with partial optimal transport. Working toward a characterisation result

Latent-Space Learning of Nonlinear Dynamics with JEPA-Style Models Completed

- Designed a testbed of nonlinear oscillators indexed by a single scalar parameter, rendered so the parameter appears only as a non-dynamical color label, to test whether latent-prediction models learn a system's dynamics rather than its appearance
- The learned equation embedding forms a smooth one-dimensional curve (>99.99% of variance in the first two principal components) and predicts waveforms of held-out equations never seen in training; what the model represents is reproducible across seeds and color palettes, while the precise embedding geometry is not

Posters and Presentations

T.A. Diac, M.C. Raportaru, A. Nicolin-Żaczek. *Training Large Language Models on a Novel Romanian Corpus*. Poster, International Balkan Workshop on Applied Physics and Materials Science (IBWAP), Constanța, Romania, July 2025.

Awards and Selections

- **CERN Student Internship Scholarship** (Nov-Dec 2023): selected as one of 12 students nationwide across Romanian physics programs, fully funded; advanced courses and seminars in theoretical particle physics
- **LLM-jp Collaboration member** (llm-jp.nii.ac.jp)

Technical Skills

Mathematics and Statistics: probability and statistics, Markov processes, Monte Carlo methods, extreme-value modelling, linear algebra, information theory, optimal transport

Machine Learning: TensorFlow, Hugging Face (Transformers, Datasets), CLIP; transformer, recurrent and convolutional architectures; tokenizer design, evaluation, quantization (llama.cpp, Ollama)

Programming and Systems: Python, C/C++, Bash, Git; Linux, data pipelines, reproducible training configurations; edge deployment on NVIDIA Jetson

Competitive Programming: over 300 C/C++ problems solved on pbinfo.ro

Languages

Romanian (native), English (fluent), Mandarin Chinese (elementary, HSK 3 level)